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ECON1050 Tutorial 5

Calculus II: Derivatives in Use

Week 7 - Semester 1 2026

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Today's Plan

- Recap: key ideas from Lecture 5
- Tutorial questions: Q1–Q5
- Further exercises (not done in tutorial)

Recap: Explicit and implicit functions

Explicit function

An explicit function writes one variable directly as a function of another:

$$y = f(x). \quad \Rightarrow y' = f'(x)$$

output input

Implicit function

Sometimes a relationship is given implicitly by an equation:

$$g(x, y) = c.$$

y'

In this case, y is treated as a differentiable function of x when we differentiate.

Recap: Implicit differentiation

Main idea

If y is defined implicitly by an equation in x and y, differentiate both sides with respect to x while treating y as a function of x.

Rules used most often

- Product rule when a term contains $x \cdot y$
- Chain rule when a term contains powers or functions of y
- Rearrangement to solve for y' after differentiating

$g(x,y) = c$

$xy + x + y = c$

$y + x \cdot y' + 1 + y' = 0$

$(x+1)y' = -1 - y$

$y' = \frac{-1-y}{x+1}$

Recap: Second derivative of an implicit function

How to find y''

- Start from the first-derivative equation.
- Differentiate again with respect to x.
- Use product rule / chain rule where needed.
- Substitute the expression for y' if needed, then solve for y''.

$$xy + x + y = c$$

$$y + x \cdot y' + 1 + y' = 0 \Leftarrow$$

$$y' + y' + x \cdot y'' + 0 + y'' = 0$$

$$(1+x)y'' = -2y'$$

$$y'' = \frac{-2}{1+x} y' = \frac{-2}{1+x} \cdot \frac{-1-y}{x+1}$$

Recap: Linear approximation

Formula

For x close to a, the linear approximation of $f(x)$ about $x = a$ is

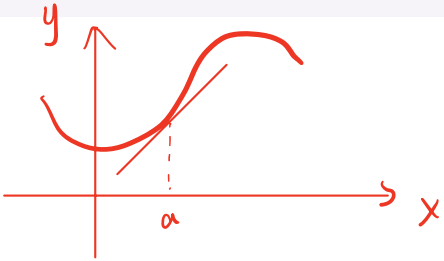
$$\boxed{f(x) \approx f(a) + f'(a)(x - a).}$$

$f(a) + f'(a)(x-a)$

tangent linear

Interpretation

The linear approximation is the tangent line to the function at $x = a$. It has the same value and the same slope as $f(x)$ at $x = a$.



Recap: Quadratic approximation

Formula

For x close to a, the quadratic approximation of $f(x)$ about $x = a$ is

$$f(x) \approx f(a) + f'(a)(x-a) + \boxed{\frac{1}{2}f''(a)(x-a)^2.}$$

$\frac{1}{2} \cdot f''(a)(x-a)^2$

Interpretation

This approximation matches the value, slope, and curvature of the original function at $x = a$.

Recap: Higher-order Taylor approximations

Taylor polynomial about $x = a$

The nth-order Taylor approximation of $f(x)$ about a is

$$f(x) \approx f(a) + \sum_{i=1}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i.$$

$i! = i \times (i-1) \times \dots \times 2 \times 1$

$f(a) + \sum_{i=1}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i$

Maclaurin polynomial

When $a = 0$, the Taylor polynomial becomes a **Maclaurin polynomial**.

~~3rd~~ 3-order Taylor approximation

$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3$

1, 2, 3

Recap: Taylor remainder and approximation error

Taylor formula with remainder

If f is differentiable $n+1$ times, then

$$f(x) = \underbrace{f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n}_{\text{n-th order Taylor approximation}} + \boxed{R_{n+1}(x)}.$$

Lagrange remainder

For some z between 0 and x,

$$R_{n+1}(x) = \frac{f^{(n+1)}(z)}{(n+1)!}x^{n+1}.$$

This gives an upper bound on the approximation error.

Recap: Elasticity

General definition

For a differentiable function $f(x)$, elasticity at x is

$y = f(x)$

$$\mathrm{El}_x f(x) = \frac{xf'(x)}{f(x)} = \frac{x}{y} \cdot \frac{dy}{dx}$$

For demand $Q(p)$, the point elasticity of demand is

$$\varepsilon(p) = \frac{pQ'(p)}{Q(p)}.$$

$$\varepsilon(p) = \frac{p}{Q(p)} \cdot Q'(p)$$

Classification

- $|\varepsilon| > 1$: elastic
- $|\varepsilon| = 1$: unit elastic
- $|\varepsilon| < 1$: inelastic

elastic

Recap: Elasticity and revenue

Revenue

If demand is $D(p)$, then revenue is

$$R(p) = pD(p). = p \cdot D$$

product rule

Elasticity-revenue relationship

The elasticity of revenue with respect to price satisfies

$$\boxed{\mathrm{El}_p R(p) = 1 + \mathrm{El}_p D(p).}$$

Hence revenue rises with price when demand is inelastic, and falls with price when demand is elastic.

Tutorial Q1

Question 1

Given

$$x - y + 3xy = 2,$$

find:

1. the first derivative y';
2. the second derivative y''.

① $x - y + 3xy = 2$

$1 - y' + 3y + 3xy' = 0 \Leftarrow$

$(-1+3x)y' = -1-3y$

$y' = \frac{-1-3y}{-1+3x}$

$y' = \boxed{\frac{1+3y}{1-3x}}$

② $1 - y' + 3y + 3xy' = 0$

$0 - y'' + 3y' + 3y' + 3xy'' = 0$

$0 - y'' + 3xy'' + 6y' = 0$

$6y' = y'' - 3xy''$

$6y' = (1-3x)y''$

$\Rightarrow \frac{6y'}{1-3x} = y''$

$\frac{6}{1-3x} \cdot \frac{1+3y}{1-3x} = y'' \Rightarrow y'' = \frac{6(1+3y)}{(1-3x)^2}$

Tutorial Q1: First and second derivatives of an implicit function

Knowledge used

- Product rule for the term $3xy$
- Implicit differentiation treating y as a function of x
- Differentiate again to obtain y''

Solution

Differentiating $x - y + 3xy = 2$ with respect to x gives

$$1 - y' + 3y + 3xy' = 0 \Rightarrow y'(-1 + 3x) = -1 - 3y \Rightarrow y' = \frac{1 + 3y}{1 - 3x}.$$

Differentiate again:

$$-y'' + 3y' + 3y' + 3xy'' = 0 \Rightarrow y''(3x - 1) = -6y' \Rightarrow y'' = \frac{6y'}{1 - 3x} = \frac{6}{1 - 3x}\left(\frac{1 + 3y}{1 - 3x}\right) = \frac{6 + 18y}{(1 - 3x)^2}.$$

Tutorial Q2

Question 2

Given

$$2x^2 + 6xy + y^2 = 18,$$

find:

1. the first derivative y';
2. the values of y' and y'' at $\boxed{(x, y) = (1, 2)}$

① $2x^2 + 6xy + y^2 = 18$

$4x + 6y + 6xy' + 2y \cdot y' = 0$

$2x + 3y + 3xy' + y \cdot y' = 0$) $\div 2$

$(y + 3x)y' = -2x - 3y$) rearrange

$y' = \dfrac{-2x - 3y}{y + 3x}$) $\div$

② $2x + 3y + 3xy' + y \cdot y' = 0$

$2 + 3y' + 3y + 3xy'' + y' \cdot y' + y \cdot y'' = 0$

$(3x + y)y'' = -2 - 3y' - 3y - (y')^2$

$y'' = \dfrac{-2 - 3y' - 3y - (y')^2}{3x + y}$

$y'' = \dfrac{-2 - 3y - 3\frac{-2x-3y}{y+3x} - \left(\frac{-2x-3y}{y+3x}\right)^2}{y + 3x}$

Plug $(x, y) = (1, 2)$ into y' and y''

Tutorial Q2: First derivative and value at $(1, 2)$

Knowledge used

- Differentiate each term with respect to x
- Use product rule for $6xy$ and chain rule for y^2
- Substitute the point after solving for y'

Solution

Differentiate: $4x + 6y + 6xy' + 2yy' = 0$. Collect the y' terms:

$$y'(6x + 2y) = -4x - 6y.$$

So

$$y' = \frac{-4x - 6y}{6x + 2y} = -\frac{2x + 3y}{3x + y}.$$

At $(x, y) = (1, 2)$,

$$y' = -\frac{2 \cdot 1 + 3 \cdot 2}{3 \cdot 1 + 2} = -\frac{8}{5}.$$

Tutorial Q2: Second derivative at $(1, 2)$

Knowledge used

- Differentiate the expression for y' again
- Use the quotient rule and then substitute the known value of y'

Solution

Starting from $y' = \boxed{-\frac{2x+3y}{3x+y}}$, we differentiate again:

Quotient rule

$$y'' = \frac{-(2+3y')(3x+y) - (-)(2x+3y)(3+y')}{(3x+y)^2} \quad \Rightarrow \quad \boxed{y'' = \frac{7y - 7xy'}{(3x+y)^2}}.$$

Substitute for y':

$$\boxed{y'' = \frac{42xy + 7y^2 + 14x^2}{(3x+y)^3}}.$$

At $(1, 2)$,

$$y'' = \frac{42(1)(2) + 7(2)^2 + 14(1)^2}{(3 \cdot 1 + 2)^3} = \frac{126}{125}.$$

Tutorial Q3

Question 3

Let

$$\boxed{f(x) = (1+x)^5.}$$

Q3 and Q4 10 mins

Find:

1. the linear approximation about $a = 0$; $f(x) \approx f(a) + f'(a)(x-a) \Rightarrow f'(x)$
2. the quadratic approximation about $a = 0$; $f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 \Rightarrow f''(x)$ $a = 0$
3. approximate values for $\boxed{1.05^5}$ using both approximations.

$f(x) = (1+x)^5$

$f'(x) = 5(1+x)^4 \cdot 1 = 5(1+x)^4$

$f''(x) = 5 \cdot 4(1+x)^3 = 20(1+x)^3$

$f(0) = (1+0)^5 = 1$

$f'(0) = 5 \cdot (1+0)^4 = 5$

$f''(0) = 20 \cdot (1+0)^3 = 20$

$f(x) \approx 1 + 5(x-0) = 1 + 5x$: linear approxi.

$f(x) \approx 1 + 5(x-0) + \frac{20}{2} \cdot (x-0)^2 = 1 + 5x + 10x^2$: Quadratic approx.

$(1.05)^5 = (1+0.05)^5 = f(0.05)$; $x = 0.05$

linear: $(1.05)^5 = f(0.05) \approx 1 + 5 \times 0.05$

Quadratic: $(1.05)^5 = f(0.05) \approx 1 + 5 \times 0.05 + 10 \times (0.05)^2$

Tutorial Q3: Linear and quadratic approximations

Knowledge used

- Linear approximation formula about $a = 0$
- Quadratic approximation formula about $a = 0$
- Substitute $x = 0.05$ because $1.05^5 = (1 + 0.05)^5$

Solution

Since

$$f(x) = (1 + x)^5, \qquad f'(x) = 5(1 + x)^4, \qquad f''(x) = 20(1 + x)^3,$$

we have

$$f(0) = 1, \qquad f'(0) = 5, \qquad f''(0) = 20.$$

So the linear approximation is $f(x) \approx 1 + 5x$, and the quadratic approximation is $f(x) \approx 1 + 5x + 10x^2$. At $x = 0.05$, $1.05^5 \approx 1 + 5(0.05) = 1.25$ from the linear approximation, and $1.05^5 \approx 1 + 5(0.05) + 10(0.05)^2 = 1.275$ from the quadratic approximation.

Tutorial Q4

Question 4

Let

$$f(x) = \sqrt{2x}.$$

Find the third-order Taylor polynomial of $f(x)$ about $a = 1$.

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3$$

$$f(x) = (2x)^{\frac{1}{2}} = \sqrt{2x}$$

$$f'(x) = \frac{1}{2}(2x)^{-\frac{1}{2}} \cdot 2 = (2x)^{-\frac{1}{2}} = \frac{1}{\sqrt{2x}}$$

$$f''(x) = -\frac{1}{2}(2x)^{-\frac{3}{2}} \cdot 2 = -(2x)^{-\frac{3}{2}} = \frac{-1}{(2x)^{\frac{3}{2}}}$$

$$f'''(x) = (-1)\cdot\left(-\frac{3}{2}\right)(2x)^{-\frac{5}{2}} \cdot 2 = 3 \cdot (2x)^{-\frac{5}{2}} = \frac{3}{(2x)^{\frac{5}{2}}}$$

$$f(a) = f(1) = \sqrt{2} \qquad f''(a) = f''(1) = \frac{-1}{(2)^{\frac{3}{2}}}$$

$$f'(a) = f'(1) = \frac{1}{\sqrt{2}} \qquad f'''(a) = f'''(1) = \frac{3}{(2)^{\frac{5}{2}}}$$

$$f(x) \approx \sqrt{2} + \frac{1}{\sqrt{2}}(x-1) + \frac{-1}{(2)^{\frac{3}{2}}} \cdot \frac{1}{2}(x-1)^2 + \frac{3}{(2)^{\frac{5}{2}}} \cdot \frac{1}{3!}(x-1)^3$$

$$= \sqrt{2} + \frac{1}{\sqrt{2}}(x-1) - \frac{1}{(2)^{\frac{5}{2}}}(x-1)^2 + \frac{1}{(2)^{\frac{7}{2}}}(x-1)^3$$

Tutorial Q4: Third-order Taylor polynomial

Knowledge used

- Third-order Taylor polynomial about $x = a$
- Compute $f(a), f'(a), f''(a), f'''(a)$ at $a = 1$

Solution

Write

$$f(x) = (2x)^{1/2}.$$

Then

$$f'(x) = (2x)^{-1/2}, \qquad f''(x) = -(2x)^{-3/2}, \qquad f'''(x) = 3(2x)^{-5/2}.$$

At $x = 1$,

$$f(1) = \sqrt{2}, \quad f'(1) = \frac{1}{\sqrt{2}}, \quad f''(1) = -\frac{1}{2\sqrt{2}}, \quad f'''(1) = \frac{3}{4\sqrt{2}}.$$

Hence the third-order Taylor polynomial is

$$\sqrt{2x} \approx \sqrt{2} + \frac{1}{\sqrt{2}}(x-1) - \frac{1}{4\sqrt{2}}(x-1)^2 + \frac{1}{8\sqrt{2}}(x-1)^3.$$

Tutorial Q5

Question 5

For each demand function, compute the elasticity and determine when demand is elastic.

(a) $\boxed{Q(p) = a - bp}$

(b) $Q(p) = p^{-\lambda}$

(c) $Q(p) = 100 - 4p^2$

$\varepsilon(p) = \frac{p}{Q(p)} Q'(p)$

$|\varepsilon(p)| \geq 1$

(a) $Q(p) = a - bp$

$Q'(p) = -b$

$\varepsilon(p) = \frac{p}{a-bp} \cdot (-b) = \frac{-bp}{a-bp}$

$\left|\frac{-bp}{a-bp}\right| \geq 1$

$\frac{bp}{a-bp} \geq 1$

$bp \geq a - bp$

$2bp \geq a$

$p \geq \frac{a}{2b}$

(b) $Q(p) = p^{-\lambda}$

$Q'(p) = -\lambda \cdot p^{-\lambda-1}$

$\varepsilon(p) = \frac{p}{p^{-\lambda}} \cdot (-\lambda \cdot p^{-1-\lambda})$

$= p^{1+\lambda-1-\lambda} \cdot (-\lambda)$

$= -\lambda$

$|-\lambda| \geq 1$

$\lambda \geq 1$

(c) $Q(p) = 100 - 4p^2$

$Q'(p) = -8p$

$\varepsilon(p) = \frac{p}{100-4p^2} \cdot (-8p) = \frac{-8p^2}{100-4p^2} = \frac{-2p^2}{25-p^2}$

$\left|\frac{-2p^2}{25-p^2}\right| \geq 1$

$100 - 4p^2 > 0$

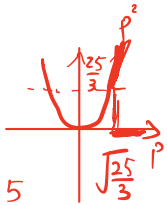
$25 - p^2 > 0$

$\frac{2p^2}{25-p^2} \geq 1$

$2p^2 \geq 25 - p^2 \Rightarrow$

$3p^2 \geq 25$

$p^2 \geq \frac{25}{3}$



$[\frac{5}{\sqrt{3}}, \infty)$

Tutorial Q5(a): Linear demand

Knowledge used

- Use $\varepsilon(p) = \dfrac{pQ'(p)}{Q(p)}$
- Compare $|\varepsilon(p)|$ with 1

Solution

If $Q(p) = a - bp$, then $Q'(p) = -b$, so

$$\varepsilon(p) = \frac{p(-b)}{a - bp} = -\frac{bp}{a - bp}.$$

Demand is elastic when

$$\left|\frac{-bp}{a - bp}\right| \geq 1 \Rightarrow bp \geq a - bp \Rightarrow 2bp \geq a.$$

Hence

$$p \geq \frac{a}{2b}.$$

Tutorial Q5(b): Isoelastic demand

Knowledge used

- Differentiate a power function
- Use the elasticity formula directly

Solution

If $Q(p) = p^{-\lambda}$, then

$$Q'(p) = -\lambda p^{-\lambda-1}.$$

So

$$\varepsilon(p) = \frac{p(-\lambda p^{-\lambda-1})}{p^{-\lambda}} = -\lambda.$$

Therefore demand is elastic for all p when

$$\lambda \geq 1.$$

Tutorial Q5(c): Quadratic demand

Knowledge used

- Differentiate the demand function first
- Solve the inequality $|\varepsilon(p)| \geq 1$

Solution

If $Q(p) = 100 - 4p^2$, then

$$Q'(p) = -8p.$$

So

$$\varepsilon(p) = \frac{p(-8p)}{100 - 4p^2} = -\frac{8p^2}{100 - 4p^2}.$$

Demand is elastic when

$$\left| \frac{-8p^2}{100 - 4p^2} \right| \geq 1 \Rightarrow 8p^2 \geq 100 - 4p^2 \Rightarrow 12p^2 \geq 100 \Rightarrow p^2 \geq \frac{25}{3}.$$

So

$$p \geq \frac{5}{\sqrt{3}}.$$

Further exercise: Q7

Knowledge used

- Linear approximation about a chosen point
- Evaluate both the function and its derivative at the base point

Solution

Let

$$f(k) = 120k^{1/3}.$$

About $k = 64$,

$$f(64) = 120(64)^{1/3} = 480, \qquad f'(k) = 40k^{-2/3}, \qquad f'(64) = 40\left(\frac{1}{16}\right) = \frac{5}{2}.$$

So the linear approximation is

$$f(k) \approx 480 + \frac{5}{2}(k - 64) = 320 + 2.5k.$$

Further exercise: Q8

Knowledge used

- Third-order Maclaurin polynomial
- Substitute a small value of x to approximate the function

Solution

For

$$f(x) = e^{-6x},$$

we have

$$f(0) = 1, \quad f'(0) = -6, \quad f''(0) = 36, \quad f'''(0) = -216.$$

Hence the third-order Maclaurin polynomial is

$$e^{-6x} \approx 1 - 6x + 18x^2 - 36x^3.$$

At $x = 0.02$,

$$e^{-0.12} = e^{-6(0.02)} \approx 1 - 6(0.02) + 18(0.02)^2 - 36(0.02)^3 = 0.886912.$$

Further exercise: Q12

Knowledge used

- Use logarithms to simplify multiplicative expressions
- Differentiate implicitly with respect to an exogenous variable

Solution

Suppose equilibrium is given by

$$AP^{-\alpha}r^{-\beta} = S, \qquad A, \alpha, \beta, S > 0.$$

Take natural logs:

$$\ln A - \alpha \ln P - \beta \ln r = \ln S.$$

Differentiate with respect to r:

$$0 - \alpha \frac{1}{P}\frac{dP}{dr} - \beta \frac{1}{r} = 0.$$

So

$$\frac{dP}{dr} = -\frac{\beta}{\alpha}\frac{P}{r} < 0.$$

Key takeaways

- Implicit differentiation treats y as a function of x and often uses the product rule or chain rule.
- Linear, quadratic, and higher-order Taylor approximations are useful when we only care about behaviour near a point.
- Elasticity measures percentage sensitivity and is especially useful for demand and revenue analysis.